



**Verified Carbon
Standard**

METHANE RECOVERY PROJECT PALMAS DE IXCAN

Project title	Methane Recovery Project Palmas de Ixcan
Project ID	Verra Project ID 5553
Crediting period	01-August-2025 to 31-July -2032
Original date of issue	For pipeline listing, 08-May-2025 is the date of submission
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Version	V01
VCS Standard Version	VCS Standard, 4.7
Prepared by	Negocios Energéticos de Occidente S.A. (NEOSA)

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1 PROJECT DETAILS

1.1 Summary Description of the Project

The Methane Recovery Project in Palmas de Ixcan Planta San Roman (referred to as “The Project”) is the implementation of an anaerobic wastewater treatment facility at the palm oil factory in Petén, Guatemala and would be a non-grouped project.

The project location is: 16.18311289755734, -90.42316906207729.

The project is expected to generate greenhouse gas (GHG) emission reductions by introducing an anaerobic digester, installing a membrane over the lagoon to capture methane emitted and in a second phase inject renewable energy into the national grid. After the digesters, the residue will be further treated through a series of aerobic processes before being irrigated to land or returned to the composting site. Biogas generated from the digesters will be captured and utilized. Biogas will be converted and used for various purposes:

- Internal use for generators.
- Excess biogas will be flared.
- Surplus of energy will be injected into the grid.

Prior to the implementation of the project activity, wastewater produced from the processing of Fresh Fruit Bunches (FFB) from Palm Oil was distributed into open anaerobic lagoons for anaerobic treatment. After the anaerobic treatment, the treated POME was discharged through land irrigation. The annual average GHG emission reductions are estimated as tCO₂e. The yearly total GHG emission reductions are estimated as 294,803 tCO₂e

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation		VCS	AENOR CONFIA, S.A.U.	0
	...			

1.3 Sectoral Scope and Project Type

Sectoral scope ¹	13,1
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¹ Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

Project activity type

Waste Handling and Disposal, Energy Industries

1.4 Project Eligibility

1.4.1 General eligibility

The project involves the implementation of a closed anaerobic wastewater treatment system and the combustion of biogas. Therefore, as per Section 2.1 of the VCS Standard, version 4.7, the project activity is included under the scope of the VCS Program and not excluded as per Table 1: Excluded Project Activities.

The project will comply with the pipeline listing process for projects under validation and seeks to complete validation within two years of the project start date in accordance with Section 3.8.1 of the VCS Standard, version 4.7.

The applied methodology, AMS-III.H.: Methane recovery in wastewater treatment – Version 19.0 is currently listed as an eligible CDM Methodology for the VCS Standard.

1.4.2 AFOLU project eligibility

N/A

1.4.3 Transfer project eligibility

N/A

1.5 Project Design

☒ Single location or installation

☐ Multiple locations or project activity instances (but not a grouped project)

☐ Grouped project

1.5.1 Grouped project design

N/A

1.6 Project Proponent

Organization name	Agroindustria Palmera San Roman S.A.
Contact person	Luis Oliva
Title	Gerente de Sostenibilidad (General Manager of Sustainability)
Address	5A. Avenida 5-55, Edificio Europlaza Torre 3, Nivel 11, Suite 1101 Z. 14 Ciudad de Guatemala, Guatemala

Telephone	+502 2328 3700
Email	loliva@pdigt.com

1.7 Other Entities Involved in the Project

Organization name	Negocios Energéticos de Occidente, S.A (NEOSA)
Role in the project	Project Developer. NEOSA elaborates and oversees the development of appropriate project design and monitoring techniques in line with the guidelines of the VCS Standard.
Contact person	Isabel Aguirre
Title	Project Coordinator
Address	5 avenida 11-70 zona 1, Guatemala City, Guatemala
Telephone	+502 56303039
Email	iaguirre@econegocios.com.gt

1.8 Ownership

The project proponent has legal rights to operate the project activity.

All Legal documents to confirm such claim will be made available to the auditors during the validation process.

1.9 Project Start Date

Project start date	01-August-2025
Justification	The official biogas plant commissioning date. As per the VCS Standard, Non-AFOLU projects shall complete validation within two years of the project start date (VCS Standard, V4.7, Section 3.8.1). Non-AFOLU projects applying a standardized method for determining additionality shall initiate the project pipeline listing process within two years of the project start date and may complete validation any time up to concurrent with the first verification (VCS Standard, V4.7, Section 3.8.7 Standardized Methods). Project applies standardized method for determining additionality, methodological tool “Positive list of

technologies". Therefore, the project shall initiate pipeline listing before 01-August-2027.

1.10 Project Crediting Period

Crediting period	☒ Seven years, twice renewable ☐ Ten years, fixed ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
Start and end date of first or fixed crediting period	01- August -2025 to 31-July-2032 (including both days).

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

☒ < 300,000 tCO₂e/year (small project)

☐ ≥ 300,000 tCO₂e/year (large project)

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
01-August-2025 to 31-July-2026	40,533
01-August-2026 to 31-July-2027	42,378 (energy injected to grid starts)
01-August-2027 to 31-July-2028	42,378
01-August-2028 to 31-July-2029	42,378
01-August-2029 to 31-July-2030	42,378
01-August-2030 to 31-July-2031	42,378
01-August-2031 to 31-July-2032	42,378
Total estimated ERRs during the first or fixed crediting period	294,803
Total number of years	7
Average annual ERRs	42,114

1.12 Description of the Project Activity

The single location will consist of the implementation of biogas recovery, flaring and/or utilization systems from industrial wastewater. The project will reduce greenhouse gas (GHG) emissions from wastewater by capturing the biogas that would otherwise have been generated and released to the atmosphere from the baseline wastewater treatment systems. The wastewater will be contained in the anaerobic lagoon digester (ALD) allowing the effluent water to make contact with the anaerobic bacteria in an anaerobic lagoon converting the organic matter into biogas.

The biogas is latter transferred to either the flare in case of surplus/emergency or to biogas processor before reaching the generator. The electricity generated will be used by the project proponent although excess electricity will be supplied to the national grid.

The project activity employs the following technologies to be implemented in the project:

- An anaerobic lagoon digester (ALD)
- Biogas collection/distribution equipment
- Flare
- Biogas generator

The material used to cover the anaerobic lagoon is a synthetic high-density polyethylene (HDPE). This covering approach in the lagoon prevents methane emissions and reduces odor. The lagoon has a volume of 15,000 m³.

To combust excess biogas the project has implemented an enclosed flare. This is used in case of emergencies if the volume of biogas exceeds the project's capacity.

The project will start with 1 biogas generator and will produce less than 5 MW. If excess energy is produced by the project, it will be supplied to the national grid.

1.13 Project Location

The project's coordinates are: 16.18311289755734, -90.42316906207729. It is located in the municipality of Sayaxché in the department of Peten, Guatemala.

1.14 Conditions Prior to Project Initiation

The project has not been implemented to generate GHG emissions for the purpose of their subsequent reduction, removal or destruction. In the conditions existing prior to the project initiation, methane from wastewater is being emitted into the atmosphere. The project's main purpose is the generation of energy for internal use.

The conditions existing prior to the project initiation are the same as the baseline scenario, please refer to Section 3. Baseline Scenario for further details.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

Name	Description
Political Constitution of the Republic of Guatemala	Supreme law of the country, regulates everything concerning the rights and obligations of people who are in the national territory, as well as the organization of the state and establishes the obligation to prevent environmental pollution and maintain ecological balance.
Protection and Improvement of the Environment Law, Decree No. 68-86	Framework law that aims to ensure the maintenance of the ecological balance and the quality of the environment and where the guidelines for environmental legal compliance are dictated
Hydrocarbon Marketing Law, Decree No. 109-97 and its Regulations Government Agreement No. 522-99	Regulates all provisions related to the marketing and storage of hydrocarbons, ensuring that safety standards, environmental protection, quality, volumes and dispatch weights are met.
Framework Law to Regulate the Reduction of Vulnerability, Mandatory Adaptation to the Effects of Climate Change, Decree No. 7-2013	Law that establishes the necessary regulations to prevent, plan and respond in an urgent, adequate, coordinated and sustained manner to the impacts of climate change.
Penal Code, Decree No. 27-73	Law that establishes the applicable penalties for what are classified as crimes or misdemeanors in the country, including penalties for crimes against the environment.
Municipal Code, Decree No. 12-2002	Code that develops the constitutional principles referring to the organization, government, administration and operation of municipalities and contains variables linked to the environment.
Health Code, Decree No. 90-97	General observance code that addresses everything related to the health sector in Guatemala.

Labor Code, Decree No. 1441	It regulates the rights and obligations of employers and workers, during work, and creates institutions to resolve their conflicts.
Evaluation, Control and Monitoring Regulations Environmental, Government Agreement No. 137-2016	Regulation that establishes the use of instruments that facilitate the evaluation, control and environmental monitoring of activities, works, industries or projects
Regulation of Discharges and Reuse of Wastewater and Sludge Disposal, Government Agreement No. 236-2006	Regulation that establishes the criteria and requirements that must be met for the discharge and reuse of wastewater, as well as the disposal of sludge. The objective of the regulation is to protect and/or recover the receiving bodies.
General Manual of the Regulation of Discharges and Reuse of Wastewater and Sludge Disposal, Ministerial Agreement No. 105-2008	Establishes the procedures for taking samples of wastewater, water for reuse and sludge; and the guide for the preparation of the Technical Study of the generating entity.
Regulations on the Registration, Marketing and Control of Agricultural Pesticides and Related Substances, Government Agreement No. 377-90	The regulation regulates the activities of registration, labelling, import, production, storage, transportation, sale and use of pesticides in general.
Occupational Health and Safety Regulation, Government Agreement No. 229-2014 and its modifications in Government Agreement No. 33-2016 (Reforms to the Occupational Health and Safety Regulation).	Its purpose is to regulate the general health and occupational safety conditions of workers in order to protect life, health and integrity.
National Policy for the Comprehensive Management of Residues and Solid Waste, Government Agreement No.281-2015	This policy aims to reduce the levels of environmental pollution produced by waste and solid waste in Guatemala.
National Policy for Environmental Management Rational of Chemical Products and Waste Dangerous in Guatemala, Government Agreement No. 341-2013	The objective of the policy is to achieve rational environmental management of chemicals and hazardous waste at the national level.

National Climate Change Policy, Agreement Government No. 329-2009.	Policy that contributes to the fulfillment of the Sustainable Development Goals, specifically Goal 13, which indicates the urgency of adopting measures to combat the effects of climate change.
Taxative List of Projects, Works, Industries or Activities, Ministerial Agreement No. 199-2016	List that classifies projects, works, industries or activities into categories according to the environmental impact they generate.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

1.18 Sustainable Development Contributions

The project contributes to the following Sustainable Development Goals.

SDG 6. Clean Water and Sanitation:	Through the improved use of water.
SDG 7. Affordable and Clean Energy:	Increasing the share of renewable energy in the national energy grid.
SDG 8. Decent work and Economic growth:	Developing policies that support productive activities and decent job creation. Improving resource efficiency in consumption and production, decoupling economic growth from environmental degradation.
SDG 9. Industry, Innovation, and Infrastructure	Upgrading infrastructure with increased resource use efficiency and adoption of clean and environmentally sound technologies and industrial processes.
SDG 12. Responsible Consumption and Production	Creation of products with sustainably managed and efficient use of natural products.
SGD 13. Climate Action	Through significant emission reduction, turning waste into energy and treating wastewater.

Guatemala has a strong commitment to all 17 UN Sustainable Development Goals, however in 2016 it prioritized its national goals². The project contributes to the following indicators recognized as Guatemalan national goals:

6.3.1	Proportion of wastewater treated safely.
6.4.1	Change in water use efficiency over time (through efficient utilization of water resources).
7.b.1	Investments in energy efficiency.
8.8.2	Increased national compliance with labor rights.

² https://pnd.gt/Documentos/ODS_Metas_priorizadas.pdf

12.2.1	Material footprints in absolute terms.
12.6.1	Number of companies that publish sustainability reports.

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

The project does not have any leakage. All the equipment installed is new.

1.19.2 Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

1.19.3 Further Information

There is not further information that could affect the projects eligibility.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification	The stakeholders were identified as workers, people of the surrounding area/villages, and the local government representatives.
Legal or customary tenure/access rights	N/A as the project is to be implemented in the palm mill and mill has all the rights over the property.
Stakeholder diversity and changes over time	The stakeholders have only one type of community as the project is located in the palm oil processing mill, workers and nearby villagers are the only affected stakeholders.
Expected changes in well-being	N/A refers to the above justification.
Location of stakeholders	Surrounding area

Location of resources	Palm Oil Processing Mill
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2.1.2 Stakeholder Consultation and Ongoing Communication

Date of stakeholder consultation	TBD
Stakeholder engagement process	
Consultation outcome	
Ongoing communication	
Stakeholder input	

2.1.3 Free Prior and Informed Consent

Obtaining consent	Consent for implementation has been granted by relevant government authorities. The property is private land.
Outcome of FPIC	Consent for implementation has been granted.

2.1.4 Grievance Redress Procedure

Development process	Any person/stakeholder will be open to raise any grievance/complaint/feedback as applicable pertaining to the VCS project.
Grievance redress procedure	If conflicts or grievances arise, the project relies on the feedback and grievance redress policy. This process is designed to be an efficient, fair and accessible mechanism for resolving complaints and conflicts in a transparent and comprehensive manner. During stakeholder consultations, stakeholders have been informed of the proper way they can make grievances, contacting the project proponent through the official email provided or by contacting the project manager. Grievances will be assessed to identify and verify the cause, actors and scale of grievances, and a resolution will be recommended based on feedback from the stakeholders. If grievances are not amicably resolved after this process, it will be submitted to an unbiased third party for a formal mediation and arbitration process,

	and subject to a hearing at which both disputing parties can testify. All cases will be referred and examined to the extent allowed by Guatemalan laws and regulations of the relevant jurisdiction before decisions are made. To ensure transparency, the project will maintain a record book with all received and resolved grievances which can be viewed upon request by project stakeholders or auditors.
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2.1.5 Public Comments

Comments received	Actions taken
TBD	

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

2.2.2 Risk Assessment

	Risks identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to stakeholders' wellbeing	Exposure to methane leaks or explosions from improperly managed capture systems Health impacts from air pollution, including hydrogen sulfide (H₂S) and other gases Water contamination affecting local communities and ecosystems Natural disasters (flooding, earthquakes, storms) compromising infrastructure.	Regular monitoring of methane capture system integrity and gas leak detection. Proper wastewater treatment and drainage to prevent contamination. Risk assessments for natural disasters and reinforcement of infrastructure.
Risks to stakeholder participation	Language and cultural barriers limiting effective participation. Resistance from stakeholders due to misinformation or distrust.	Establishment of a multi-stakeholder meeting for participation and dialogue. Translation of key project information into local languages and use of culturally appropriate communication methods if necessary. Active engagement through community meetings, surveys, and participatory decision-making.

		Transparency in project implementation.
Working conditions	Working conditions shall be as per the applicable laws. Possible risks are related to the nature of the work: Exposure to hazardous gases and chemicals	Compliance with national labor laws. Personal protective equipment (PPE) and regular safety training for all workers. Monitoring and reporting mechanisms for workplace conditions and grievances.
Safety of women and girls	Discrimination in employment opportunities and wages.	Policies promoting equal pay and opportunities for women.
Safety of minority and marginalized groups, including children	Project complies with all applicable national laws and therefore does not hire minors. Project has commitment to provide equal opportunities to minorities and marginalized groups.	N/A
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	Air pollution from methane leaks, flaring, or improper gas handling Noise pollution from machinery used in methane capture and palm oil processing Water pollution due to untreated wastewater discharge	Use of best available technologies for methane capture efficiency and leak prevention. Installation of noise-reducing barriers and scheduled maintenance of machinery. Implementation of strict wastewater treatment protocols before discharge. Waste management systems for proper disposal and recycling of hazardous materials

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

	Risks identified ³	Mitigation or preventative measure(s) taken
Discrimination	N/A. Project has strong nondiscrimination policies.	Transparent recruitment and hiring processes. Establishment of grievance mechanism.
Sexual harassment	Harassment of female workers by supervisors, co-workers, or third parties.	Development and enforcement of a zero-tolerance sexual harassment policy. Establishment of grievance mechanism.

³ The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

Equal pay for equal work	Wage disparities between men and women or local and migrant workers.	Enforcement of equal pay policies
Gender equity in labor and work	N/A. Workers are hired on a merit based method, while respecting the cultural norms of the region.	
Forced labor	N/A. Project does not engage in forced labor.	
Child labor	N/A. Project complies with all local laws and does not engage with the hiring of minors.	
Human trafficking	N/A. Project complies with all local laws and does not engage with human trafficking.	

2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	The project takes place in private property and does not impact the rights of IPs, LCs, and customary rights holders.

2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Mitigation(s) or preventative measure taken
No risk identified	The project takes place in private property and does not impact indigenous peoples and their cultural heritage.

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	Project activities take place on private property

2.3.5 Benefit Sharing

Process used to design the benefit sharing plan	Not applicable, the project is waste water treatment project and biogas recovery and no stakeholders are affected or have legal right to land or project.
Summary of the benefit sharing plan	Not Applicable

Approval and dissemination of benefit sharing plan

Not Applicable

2.4 Ecosystem Health

Project takes place on existing infrastructure and will not impact ecosystem health in a negative way. Project activities are expected to be net positive.

	Risks identified	Mitigation or preventative measure(s) taken
Impacts on biodiversity and ecosystems	N/A	
Soil degradation and soil erosion	N/A	
Water consumption and stress	N/A	

2.4.1 Rare, Threatened, and Endangered Species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes

☒ No

2.4.2 Introduction of Species

N/A

2.4.3 Ecosystem Conversion

N/A

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	AMS-III.H	AMS-III.H Methane recovery in wastewater treatment	19

Methodology	AMS-I.D	Grid connected renewable electricity generation	18
Tool	Tool 03	Calculate project or leakage CO2 emissions from fossil fuel combustion	03.0
Tool	Tool 05	Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation	03.0
Tool	Tool 06	<i>Project Emissions from Flaring</i>	04.0
Tool	Tool 07	Tool to calculate the emission factor for an electricity system	07.0
Tool	Tool 08	Tool to determine the mass flow of a greenhouse gas in a gaseous stream	03.0
Tool	Tool 32	<i>Positive list of technologies</i>	04.0

3.2 Applicability of Methodology

Methodology ID	Applicability condition	Justification of compliance
AMS-III.H	This methodology comprises measures that recover biogas from biogenic organic matter in wastewater by means of one, or a combination, of the following options: (a) Substitution of aerobic wastewater or sludge treatment systems with anaerobic systems with biogas recovery and combustion; (b) Introduction of anaerobic sludge treatment system with biogas recovery and combustion to a wastewater treatment plant without sludge treatment; (c) Introduction of biogas recovery and combustion to a sludge treatment system; (d) Introduction of biogas recovery and combustion to an anaerobic wastewater treatment system such as anaerobic reactor, lagoon, septic tank or an on-site industrial plant. (e) Introduction of anaerobic wastewater treatment with	Applicable: Option D. The project activity consists of covering four anaerobic lagoons to recover the biogas for combustion.

	biogas recovery and combustion, with or without anaerobic sludge treatment, to an untreated wastewater stream; (f) Introduction of a sequential stage of wastewater treatment with biogas recovery and combustion, with or without sludge treatment, to an anaerobic wastewater treatment system without biogas recovery (e.g. introduction of treatment in an anaerobic reactor with biogas recovery as a sequential treatment step for the wastewater that is presently being treated in an anaerobic lagoon with.	
AMS-III.H	In cases where baseline system is anaerobic lagoon the methodology is applicable if: (a) The lagoons are ponds with a depth greater than two meters, without aeration. The value for depth is obtained from engineering design documents, or through direct measurement, or by dividing the surface area by the total volume. If the lagoon filling level varies seasonally, the average of the highest and lowest levels may be taken; (b) Ambient temperature above 15 °C, at least during part of the year, on a monthly average basis; (c) The minimum interval between two consecutive sludge removal events shall be 30 days.	Applicable: (a) The lagoons have a depth greater than two meters. (b) Ambient temperature is above 15 °C at least during part of the year, on a monthly average basis. (c) Project activities do not involve sludge treatment.
AMS-III.H	The recovered biogas from the above measures may also be utilized for the following applications instead of combustion/flaring: (a) Thermal or mechanical, electrical energy generation directly; (b) Thermal or mechanical,	Applicable: Option A. Biogas will supply generator

	electrical energy generation after bottling of upgraded biogas, in this case additional guidance provided in the appendix shall be followed; or (c) Thermal or mechanical, electrical energy generation after upgrading and distribution, in this case additional guidance provided in the appendix shall be followed: (i) Upgrading and injection of biogas into a natural gas distribution grid with no significant transmission constraints; (ii) Upgrading and transportation of biogas via a dedicated piped network to a group of end users; or (iii) Upgrading and transportation of biogas (e.g. by trucks) to distribution points for end users; (d) Hydrogen production; (e) Use as fuel in transportation applications after upgrading.	
AMS-III.H	For validation and verification of CDM projects and program of activities by a designated operation entity (DOE) using this methodology: (a) If the recovered biogas from the waste water treatment plant is only flared and not used for any other purpose, then sectoral scope13 is mandatory; (b) If the recovered biogas is used for any other purpose, then sectoral scope 13 and sectoral scope 1 are mandatory.	Applicable: Option B. Recovered gas is flared and used for electricity generation, therefore sectoral scope 13 and 1 are mandatory.
AMS-I.D	This methodology is applicable to project activities that: (a) Install a Greenfield plant; (b) Involve a capacity addition in (an) existing plant(s); (c) Involve a retrofit of (an) existing plant(s);	Applicable: Option A. Greenfield plant is used.

	(d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s).	
AMS-I.D	If the new unit has both renewable and non-renewable components (e.g. a wind/diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15 MW.	Applicable: The project will produce less than 5 MW.
AMS-I.D	Combined heat and power (co-generation) systems are not eligible under this category.	Applicable: Project will only generate power.
AMS-I.D	In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct ¹ from the existing units.	Applicable: Project will produce less than 5 MW.
AMS-I.D	In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project, the total output of the retrofitted, rehabilitated or replacement power plant/unit shall not exceed the limit of 15 MW.	Applicable: Project will produce less than 5 MW.
AMS-I.D	In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid then the	Applicable: The recovered methane will not be used for heat generation.

	baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as “AMS-I.C.: Thermal energy production with or without electricity” shall be explored.	
AMS-I.D	In case biomass is sourced from dedicated plantations, the applicability criteria in the tool “Project emissions from cultivation of biomass” shall apply.	Applicable: Biomass is not used to generate electricity.
Tool 32	The existing treatment system is an anaerobic lagoon and the wastewater discharged meets the host country regulation;	The baseline scenario of the project complies with Government Agreement No. 236-2006, Regulation of Discharges and Reuse of Wastewater and Sludge Disposal and Ministerial Agreement No. 105-2008, General Manual of the Regulation of Discharges and Reuse of Wastewater and Sludge Disposal.
Tool 32	There is no regulation in the host country that requires the management of biogas from domestic, industrial and agricultural sites;	There is no regulation in Guatemala that requires the management of biogas.
Tool 32	There is no capacity increase in the wastewater treatment system;	The wastewater treatment system will maintain the same capacity.
Tool 32	No other alternative economic activity is expected to be undertaken on the land of the existing lagoon;	No alternative economic activity will be undertaken on the land of the existing lagoon as it will continue to be used.
Tool 32	The biogas is used to generate electricity in one or more power plants, and the total nameplate capacity is below 5 MW.	The expected installed capacity is less than 5 MW

3.3 Project Boundary

Source		Gas	Included?	Justification/Explanation
Baseline	Wastewater Treatment Processes	CO ₂	No	CO ₂ emissions from the decomposition of organic waste are not accounted for
		CH ₄	Yes	Main source of emission in the baseline.
		N ₂ O	No	Excluded for simplification
	Electricity Consumption/ Generation	CO ₂	Yes	Major source of emissions in baseline.
		CH ₄	No	Excluded for simplification
		N ₂ O	No	Excluded for simplification
Project	Wastewater Treatment Processes	CO ₂	No	CO ₂ emissions from the decomposition of organic waste are not accounted for
		CH ₄	Yes	Main emission source.
		N ₂ O	No	Excluded for simplification
	Electricity Consumption/ Generation	CO ₂	Yes	Important emission source
		CH ₄	No	Excluded for simplification
		N ₂ O	No	Excluded for simplification

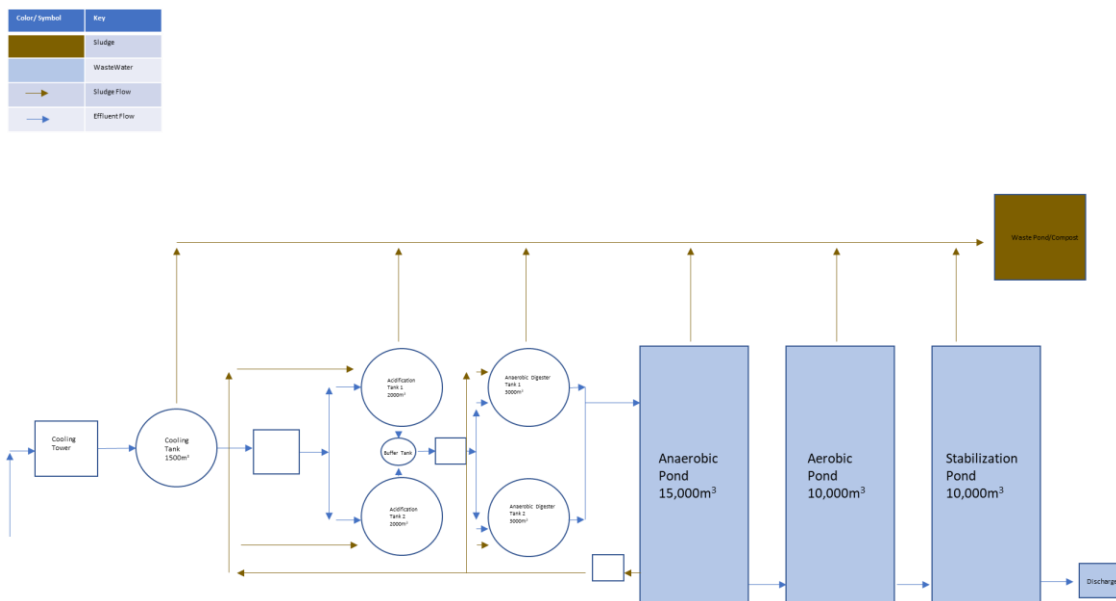
3.4 Baseline Scenario

Wastewater from the palm oil mill is characterized by a high chemical oxygen demand (COD), stemming from organic content from the palm oil production process. In the Baseline Scenario, this wastewater is treated in a lagoon system. All lagoons of this system are each deeper than 2m and the average atmospheric temperature in the region is above 15 degree Celsius⁴. These conditions create an anaerobic environment, resulting in methane generation from the breakdown of the organic content of the wastewater (where organic content of the wastewater is characterized by COD) as the wastewater progresses through the treatment ponds. Once the wastewater has completed processing through the baseline treatment system, this wastewater is irrigated as fertilizer, compliant with the legislation of Guatemala.

The sludge is removed by a system of floating pumps and sent for treatment to a composter. This removal is carried out every week.

⁴ Guatemala, Average Mean Surface Air Temperature 1990 – 2020, available online at: <https://climateknowledgeportal.worldbank.org/country/guatemala/climate-data-historical>

In the absence of project activity, the baseline scenario will be the continuation of the present wastewater treatment, including ongoing methane emissions to the atmosphere from the anaerobic lagoons.



3.5 Additionality

3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country

☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes

☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes

☒ No

Project activities are not mandated by any law, statute, or other regulatory framework.

3.5.2 Additionality Methods

Additionality is demonstrated according to methodology AMS-III.H version 19.0. The simplified procedure to demonstrate additionality can be found in the methodological tool 32, "Positive lists of technologies; version 4.0." While applying this tool, it is not required to demonstrate compliance with conditions 12 (c) and 12 (d).

The proposed project is in conformance with the positive list as demonstrated in section 3.2 Applicability of Methodology above.

Verra issued a Clarification to Tool 32 on March 18, 2025, clarifying that the tool is active for use in the VCS Program until September 30, 2025⁵.

3.6 Methodology Deviations

No methodological deviations are expected to be applied.

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

According to methodology AMS-III.H version 19.0, baseline emissions for the systems affected by the project activity may consist of:

- (a) Emissions on account of electricity or fossil fuel used ($BE_{power,y}$);
- (b) Methane emissions from baseline wastewater treatment systems ($BE_{ww,treatment,y}$);
- (c) Methane emissions from baseline sludge treatment systems ($BE_{s,treatment,y}$);
- (d) Methane emissions on account of inefficiencies in the baseline wastewater treatment systems and presence of degradable organic carbon in the treated wastewater discharged into river/ lake/ sea ($BE_{ww,discharge,y}$);
- (e) Methane emissions from the decay of the final sludge generated by the baseline treatment systems ($BE_{s,final,y}$).

$$BE_y = \{BE_{power,y} + BE_{ww,treatment,y} + BE_{s,treatment,y} + BE_{ww,discharge,y} + BE_{s,final,y}\} \quad \text{Equation (1)}$$

Where:

BE_y	Baseline emissions in year y (tCO ₂ e)
$BE_{power,y}$	Baseline emissions from electricity or fuel consumption in year y (tCO ₂ e)
$BE_{ww,treatment,y}$	Baseline emissions of the wastewater treatment systems affected by the project activity in year y (tCO ₂ e)
$BE_{s,treatment,y}$	Baseline emissions of the sludge treatment systems affected by the project activity in year y (tCO ₂ e)
$BE_{ww,discharge,y}$	Baseline methane emissions from degradable organic carbon in treated wastewater discharged into sea/river/lake in year y (tCO ₂ e).

⁵ <https://verra.org/methodologies/cdm-tool32-positive-lists-of-technologies/>

$BE_{s,final,y}$ Baseline methane emissions from anaerobic decay of the final sludge produced in year y (tCO_2e). If the sludge is controlled combusted, disposed in a landfill with biogas recovery, or used for soil application in the baseline scenario, this term shall be neglected

Project Circumstance:

- There is no sludge treatment in the baseline and in the project.
- The final discharge will not be affected by the project.
- The electric consumption of wastewater treatment is the same in the baseline and project situation.

The only system affected is the baseline wastewater treatment system. Therefore, equation (1) can be reduced to:

$$BE_y = BE_{ww,treatment,y}$$

Methane emissions from the baseline wastewater treatment systems affected by the project ($BE_{ww,treatment,y}$) are determined using the COD removal efficiency of the baseline as follows:

$$BE_{ww,treatment,y} = \sum_i (Q_{ww,i,y} * COD_{inflow,i,y} * \eta_{COD,BL,i} * MCF_{ww,treatment,BL,i}) B_{o,ww} * UF_{BL} * GWP_{CH4} \quad \text{Equation(2)}$$

Where:

$Q_{ww,i,y}$	Volume of wastewater treated in baseline wastewater treatment system i in year y (m^3). For ex ante estimation, forecasted wastewater generation volume or the designed capacity of the wastewater treatment facility can be used. However, the ex post emissions reduction calculation shall be based on the actual monitored volume of treated wastewater
$COD_{inflow,i,y}$	Chemical oxygen demand of the wastewater inflow to the baseline treatment system i in year y (t/m^3). Average value may be used through sampling with the confidence/precision level 90/10. Monitored values of the COD inflow during crediting period will be used to calculate the baseline emissions ex post
$\eta_{COD,BL,i}$	COD removal efficiency of the baseline treatment system i
$MCF_{ww,treatment,BL,i}$	Methane correction factor for baseline wastewater treatment systems i . MCF value is identified as 0.8 corresponding to anaerobic deep lagoon (depth more than 2 meters) (IPCC and methodology) i Index for baseline wastewater treatment system
$B_{o,ww}$	Methane producing capacity of the wastewater (IPCC value of 0.25 $kgCH_4/kg$ COD)
UF_{BL}	Model correction factor to account for model uncertainties (0.89)
GWP_{CH4}	Global Warming Potential for methane

In determining baseline emissions using equation (1), historical records of at least one year prior to the project implementation shall be used.

Calculation of $BE_{ww,treatment,y}$

Applying the following values for ex-ante calculations:

Parameter	Value	Source
$Q_{ww,i,y}$	127,064 m ³	Based on one y year historical data of the plant
$COD_{in,flow,i,y}$	0.068534525 m ³	Based on one year historical data of the plant
$N_{COD,BL,i}$	97%	Calculated
$MCF_{ww,treatment,i}$	0.8	Corresponding to anaerobic deep lagoon (depth more than 2 meters)
$B_{o,ww}$	0.25 kgCH ₄ /kg COD	Default value
UF_{BL}	0.89	Default value
GWP_{CH_4}	29 tCO ₂ /tCH ₄	IPCC default value

$$BE_{ww,treatment,y} = tCO_2e .$$

In accordance with applicability of conditional sectoral scopes, since the project uses recovered biogas, sectoral scope 13 and sectoral scope 1 are mandatory.

Project applies methodology AMS-I.D version 18.0, for the use of energy supplied to grid.

According to methodology AMS-I.D version 18.0, baseline emissions include only CO₂ emissions from electricity generation in power plants that are displaced due to the project activity.

$$BE_y = EG_{PJ,y} \times EF_{grid,y} \quad \text{Equation (1)}$$

Where:

$$BE_y = \text{Baseline emissions in year y (t CO}_2\text{)}$$

$$EG_{PJ,y} = \text{Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh)}$$

$$EF_{grid,y} = \text{Combined margin CO}_2\text{ emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system" (t CO}_2\text{/MWh)}$$

If the project activity is the installation of a greenfield power plant, then:

$$EG_{PJ,y} = EG_{PJ,facility,y} \quad \text{Equation (2)}$$

Where:

$$EG_{PJ,y} = \text{Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh)}$$

$EG_{PJ, facility, y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh)

Calculation of $BE_{grid, y}$

Applying the following values for ex-ante calculations:

Parameter	Value	Source
$EG_{grid, y}$	5,457 MWh	Estimated net quantity of electricity generated in year y with biogas from the new anaerobic biodigester.
$EF_{grid, y}$	0.338 tCO ₂ e /MWh	Calculated using Tool 07 Tool to calculate the emission factor for an electricity system.

$$BE_{grid, y} = 1,844 \text{ tCO}_2\text{e}$$

4.2 Project Emissions

According to the methodology AMS-III.H., project emissions may consist of:

- (a) CO₂ emissions from electricity and fuel used by the project facilities ($PE_{power, y}$),
- (b) Methane emissions from wastewater treatment systems affected by the project activity, and not equipped with biogas recovery in the project scenario ($PE_{ww, treatment, y}$),
- (c) Methane emissions from sludge treatment systems affected by the project activity, and not equipped with biogas recovery in the project situation ($PE_{s, treatment, y}$),
- (d) Methane emissions on account of inefficiency of the project activity wastewater treatment systems and presence of degradable organic carbon in treated wastewater ($PE_{ww, discharge, y}$),
- (e) Methane emissions from the decay of the final sludge generated by the project activity treatment systems ($PE_{s, final, y}$),
- (f) Methane fugitive emissions due to inefficiencies in capture systems ($PE_{fugitive, y}$),
- (g) Methane emissions due to incomplete flaring ($PE_{flaring, y}$),
- (h) Methane emissions from biomass stored under anaerobic conditions which would not have occurred in the baseline situation ($PE_{biomass, y}$)

The project activity does not involve project emissions from (c) (d) (e) (h).

$PE_{ww, treatment, y}$ is not applicable due to the wastewater treatment system being equipped with biogas recovery in the project scenario

Thus, project emissions consist of:

$$PE_y = PE_{power,y} + PE_{fugitive,y} + PE_{flaring,y} \quad \text{Equation (8)}$$

Where:

PE_y	Project activity emissions in the year y (t CO ₂ e)
$PE_{power,y}$	Emissions from electricity or fuel consumption in the year y (t CO ₂ e).
$PE_{fugitive,y}$	Methane emissions from biogas release in capture systems in year y (tCO ₂ e)
$PE_{flaring,y}$	Methane emissions due to incomplete flaring in year y (tCO ₂ e). For ex-ante estimation, baseline emission calculation for wastewater treatment (i.e. equation (2)) can be used but without the consideration of GWP for CH ₄ . However, the expost emission reduction shall be calculated as per methodological tool "Project emissions from flaring"

Calculation of $PE_{power,y}$

Electricity consumed for wastewater treatment system will not be affected by the proposed project activity, it will be the same in the baseline and in the project scenario. Therefore, for ex-ante calculations, $PE_{power,y}$ is assumed to be zero.

$$PE_{power,y} = 0 \text{ tCO}_2\text{e}$$

Equations used for the calculation of $PE_{fugitive,y}$

Project activity emissions from methane release in capture systems are determined as follows:
(a) Based on the methane emission potential of wastewater:

Calculation of $PE_{fugitive,y}$

Project activity emissions from methane release in capture system are determined as follows:

$$PE_{fugitive,y} = PE_{fugitive,ww,y} + PE_{fugitive,s,y} \quad \text{Equation (9)}$$

Where:

$PE_{fugitive,ww,y}$	Fugitive emissions through capture inefficiencies in the anaerobic wastewater treatment systems in the year y (tCO ₂ e)
$PE_{fugitive,s,y}$	Fugitive emissions through capture inefficiencies in the anaerobic sludge treatment systems in the year y (tCO ₂ e)

Since sludge treatment in the baseline scenario and the project scenario is the same, the equation can be simplified to:

$$PE_{fugitive,y} = PE_{fugitive,ww,y}$$

Where,

$$PE_{fugitive,ww,y} = (1 - CFE_{ww}) \times MEP_{ww,treatment,y} \times GWP_{CH_4} \quad \text{Equation (10)}$$

Where:

CFE_{ww}	Capture efficiency of the biogas recovery equipment in the wastewater treatment systems (a default value of 0.9 shall be used)
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$MEP_{ww,treatment,y}$ Methane emission potential of wastewater treatment systems equipped with biogas recovery system in year y (t)

$$MEP_{ww,treatment,y} = Q_{ww,y} \times B_{o,ww} * UF_{PJ} * \sum k COD_{removed,PJ,k,y} * MCF_{ww,treatment,PJ,k} \quad \text{Equation (11)}$$

Where:

$COD_{removed,PJ,k,y}$ Chemical oxygen demand removed by the treatment system k of the project activity equipped with biogas recovery in the year y (t/m³)

$MCF_{ww,treatment,PJ,k}$ Methane correction factor for the project wastewater treatment system k equipped with biogas recovery equipment (MCF value 0.8)

UF_{PJ} Model correction factor to account for model uncertainties (1.12)

Applying the following values:

CFE_{ww} 0.9

$MEP_{ww,treatment,y}$ Calculated as tCH₄

$$MEP_{ww,treatment,y} = Q_{ww,y} \times B_{o,ww} * UF_{PJ} * \sum k COD_{removed,PJ,k,y} * MCF_{ww,treatment,PJ,k}$$

Where:

Parameter	Value	Source
$Q_{ww,y}$	127,064 m ³	Based on one-year historical data of the plant
$B_{o,ww}$	0.25 kg CH ₄ /kg COD	Default value
UF_{PJ}	1.12	Default value
$COD_{removed,PJ,k,y}$	.002099744 /m ³	Calculated
$MCF_{ww,treatment,PJ,k}$	0.8	Corresponding to anaerobic deep lagoon (depth more than 2 meters)

$COD_{removed,PJ,k,y}$ is the difference between $COD_{inflow,PJ,y}$ and $COD_{outflow,PJ,y}$ of the treatment system installed in the Project.

Inflow and outflow of COD are regularly measured and monitored by project proponent. The values applied for ex-ante calculations of project emissions are the average values of COD_{inflow} and $COD_{outflow}$ for the year 2023.

Parameter	Result
$COD_{inflow,PJ,y}$	.068534525 t/m ³
$COD_{outflow,PJ,y}$	.066434781 t/m ³
$COD_{removed,PJ,k,y}$	. 002099744 t/m ³

$$PE_{fugitive,ww,y} = (1 - 0.9) * 59.76 * 29 = 137.3tCO_2$$

(b) Optionally a default value of 0.05 m³ biogas leaked/m³ biogas produced may be used as an alternative to calculations per equation (9).

Calculation of $PE_{flare,y}$

Methane emissions due to the incomplete flaring can use baseline emission calculations for wastewater and/or sludge treatment without the consideration of GWP and CH_4 for ex ante estimation. However, ex post emission reductions shall be calculated as per methodological tool 06 “Project emissions from flaring”:

(a) Based on emission potential of flaring the residual gas.

$$PE_{flare,y} = GWP_{CH_4} * \sum_{h=1}^{8760} F_{CH_4, RG, m} * (1 - n_{flare, h}) * 10^{-3} \quad \text{Equation (15)}$$

Where:

$PE_{flare,y}$	Project emissions from flaring of the residual gas in year y (tCO _{2e})
$F_{CH_4, RG, m}$	Mass flow rate of methane in the residual gas in the minute m (kg)
$n_{flare, h}$	Flare efficiency in hour h (closed flare)
GWP_{CH_4}	Global Warming Potential of methane valid for the commitment period (tCO _{2e} / tCH ₄)

To determine mass flow rate of methane in the residual gas ($F_{CH_4, RG, m}$) the following equation from Tool 08 is used:

$$F_{RG, h} = FV_{RG, h} * fV_{CH_4, RG, h} * P_{CH_4, n} \quad \text{Equation (9)}$$

Where:

$TM_{RG, h}$	Mass flow rate of methane in the residual gas in the hour h
$FV_{RG, h}$	Volumetric flow rate of the residual gas in dry basis at normal (NTP) conditions in the hour h (assume 8760h of operation)
$fV_{CH_4, RG, h}$	Volumetric fraction of methane in the residual gas on dry basis in hour h
$P_{CH_4, n}$	Density of methane at normal conditions

Parameter	Value	Source
$FV_{RG, h}$	262.41 m ³ /h	Calculated
$fV_{CH_4, RG, h}$	0.6	Industry average
$P_{CH_4, n}$	0.717 kg/m ³	CDM default data

Therefore, $F_{RG, h} = 271.42 * 0.6 * 0.717 = 112.887 \text{ kg/h}$

$$PE_{flare,y} = \sum_{h=1}^{8760} TM_{RG, h} * (1 - n_{flare, h}) * \frac{GWP_{CH_4}}{1000}$$

Parameter	Value	Source
$TM_{RG, h}$	988,897.14 kg	Mass flow rate of methane in residual gas
$n_{flare, h}$	0.9 (%)	Default value
GWP_{CH_4}	29 tCO _{2e} /tCH ₄	IPCC Value

$$PE_{flare,y} = 988,897.14 * (1 - 0.9) * 29 / 1000 = 2,867.80 \text{ tCO}_{2e}$$

According to the methodology AMS-I.D., for this type of renewable energy product activity $PE_y = 0$

4.3 Leakage Emissions

All installed technology will be new. The project does not have leakage.

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

According to the methodology AMS-III.H., the estimated GHG emission reduction are the following:

$$ER_{y,ex-ante} = BE_{y,ex-ante} - (PE_{y,ex-ante} + LE_{y,ex-ante}) \quad \text{Equation (14)}$$

Where:

$ER_{y,ex-ante}$	Ex ante emission reduction in year y (tCO ₂ e)
$BE_{y,ex-ante}$	Ex ante baseline emissions in year y (tCO ₂ e)
$PE_{y,ex-ante}$	Ex ante project emissions in year y (tCO ₂ e)
$LE_{y,ex-ante}$	There is no leakage

Parameter	Value
$ER_{y,ex-ante}$	40,533.75 tCO ₂ e
$BE_{y,ex-ante}$	43,574.86 tCO ₂ e
$PE_{y,ex-ante}$	3,041.12 tCO ₂ e
$LE_{y,ex-ante}$	0 tCO ₂ e

$$ER_{y,ex-ante} = 43,574.86 \text{ tCO}_2\text{e} - (3,041.12 \text{ tCO}_2\text{e} + 0 \text{ tCO}_2\text{e}) = 40,533.75 \text{ tCO}_2\text{e}$$

According to the methodology AMS-III.H. ex post emission reductions shall be based on the lowest value of the following:

- The amount of biogas recovered and fueled or flared (MD_y) during the crediting period, that is monitored ex post;
- Ex post calculated baseline, project and leakage emissions based on actual monitored data for the project activity;

It is possible that the project activity involves wastewater treatment systems with higher methane conversion factors (MCF) or with higher efficiency than in the baseline situation. Therefore, the emission reductions achieved by the project activity is limited to the ex post calculated baseline emissions minus project emissions using the monitored data for the project activity. The emission reductions achieved in any year are the lowest value of the following:

$$ER_{y,expost} = \min [(BE_{y,expost} - PE_{y,expost}), (MD_y - PE_{power,y})] \quad \text{Equation (15)}$$

In the case of flaring/combustion MD_y will be measured using the condition of the flaring process:

$$MD_y = BG_{burnt,y} * w_{CH_4} * DCH_4 * FE * GWP_{CH_4} \quad \text{Equation (16)}$$

Where:

$BG_{burnt,y}$	Biogas flared/combusted in year y (m ³)
w_{CH_4}	Methane content of the biogas in the year y (volume fraction)

D_{CH_4}	Density of methane at the temperature and pressure of the biogas in the year y (t/m^3)
FE	Flare efficiency in year y (fraction). If the biogas is combusted for gainful purposes, e.g. fed to an engine, an efficiency of 100 per cent may be applied.

According to the methodology AMS-I.D., the estimated GHG emission reduction are the following:

$$ER_y = BE_y - PE_y - LE_y \quad \text{Equation (21)}$$

Where:

ER_y	Emission reductions in year y (tCO_2e)
BE_y	Baseline emissions in year y (tCO_2e)
PE_y	Project emissions in year y (tCO_2e)
LE_y	Leakage emissions in year y (tCO_2e)

Parameter	Value
$ER_{y,ex-ante}$	1,844.63 tCO_2e
$BE_{y,ex-ante}$	1,844.63 tCO_2e
$PE_{y,ex-ante}$	0 tCO_2e
$LE_{y,ex-ante}$	0 tCO_2e

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	$MCF_{ww,treatment,BL,i}$
Data unit	-
Description	Methane correction factor for baseline wastewater treatment systems (Anaerobic deep lagoon - depth more than 2 meters)
Source of data	MCF value as per table 2. IPCC default values for Methane Correction Factor (MCF)
Value applied	0.8
Justification of choice of data or description of measurement methods and procedures applied	Anaerobic deep lagoon - depth more than 2 meters
Purpose of data	Calculation of baseline emission

Comments	-
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Data / Parameter	MCF _{ww,BL,discharge}
Data unit	-
Description	Methane correction factor for baseline wastewater treatment systems (land application)
Source of data	MCF value as per table 2, AMS-III. H (Version 19)
Value applied	
Justification of choice of data or description of measurement methods and procedures applied	Discharge of wastewater as fertilizer
Purpose of data	Calculation of baseline emission
Comments	-

Data / Parameter	B _{0,ww}
Data unit	Kg CH ₄ /kg COD
Description	Methane producing capacity of the wastewater
Source of data	IPCC default value as per methodology AMS-III. H (Version 19)
Value applied	0.25
Justification of choice of data or description of measurement methods and procedures applied	Default value
Purpose of data	Calculation of baseline and project emissions
Comments	-

Data / Parameter	UF _{BL}
Data unit	-
Description	Model correction factor to account for model uncertainties

Source of data	Default value as per methodology AMS-III. H (Version 19)
Value applied	0.89
Justification of choice of data or description of measurement methods and procedures applied	Default value
Purpose of data	Calculation of baseline emissions
Comments	

Data / Parameter	GWP _{CH4}
Data unit	tCO ₂ /tCH ₄
Description	Global Warming Potential for methane
Source of data	IPCC global warming potential for CH ₄ valid for the relevant commitment period
Value applied	29
Justification of choice of data or description of measurement methods and procedures applied	As per VCS Standard V4.5 For reductions and removals occurring on or after 1 January 2021, all ex-ante estimates and ex-post calculations shall be converted to CO ₂ e using GWP values from the IPCC Fifth Assessment Report. See Table 2 of VCS Standard V4.5.
Purpose of data	Calculation of baseline and project emissions
Comments	

Data / Parameter	MCF _{ww,treatment,BL,i}
Data unit	-
Description	Methane correction factor for baseline treatment systems j
Source of data	MCF value as per table 2, AMS-III. H (Version 19)
Value applied	
Justification of choice of data or description of measurement methods and procedures applied	The baseline wastewater treatment system consists of a wastewater system without biogas recovery that is discharged to lagoons with a of depth more than 2 meters, therefore the MCF value of 0.8 is chosen.
Purpose of data	Calculation of project emission

Comments	
Data / Parameter	$MCF_{ww,treatment,BL,i}$
Data unit	-
Description	Methane correction factor for baseline treatment systems j
Source of data	MCF value as per table 2, AMS-III. H (Version 19)
Value applied	
Justification of choice of data or description of measurement methods and procedures applied	The baseline wastewater treatment system consists of a wastewater system without biogas recovery in lagoons with a depth more than 2 meters, therefore the MCF value of 0.8 is chosen.
Purpose of data	Calculation of project emission
Comments	
Data / Parameter	$MCF_{ww,BL,discharge,}$
Data unit	-
Description	Methane correction factor based on the discharge pathway of the wastewater in the project scenario
Source of data	MCF value as per table 2, AMS-III. H (Version 19)
Value applied	
Justification of choice of data or description of measurement methods and procedures applied	The baseline discharge pathway of wastewater in the baseline situation is land application.
Purpose of data	Calculation of project emissions.
Comments	-
Data / Parameter	$MCF_{ww,treatment,PJ,k}$
Data unit	-
Description	Methane correction factor for wastewater treatment systems k equipped with biogas recovery equipment
Source of data	MCF value as per table 2, AMS-III. H (Version 19)

Value applied	
Justification of choice of data or description of measurement methods and procedures applied	The project wastewater treatment system consists of a wastewater system with biogas recovery in lagoons with a depth more than 2 meters, therefore the MCF value of 0.8 is chosen.
Purpose of data	Calculation of project emission
Comments	

Data / Parameter	UF _{PJ}
Data unit	-
Description	Model correction factor to account for model uncertainties
Source of data	Default value as per methodology AMS-III. H (Version 19)
Value applied	1.12
Justification of choice of data or description of measurement methods and procedures applied	Correction factor for uncertainties
Purpose of data	Calculation of project emissions.
Comments	-

Data / Parameter	CFE _{ww}
Data unit	Fraction
Description	Capture efficiency of the biogas recovery equipment in the wastewater treatment systems
Source of data	Default value as per methodology AMS-III. H (Version 19)
Value applied	0.9
Justification of choice of data or description of measurement methods and procedures applied	Default value as per methodology AMS-III. H (Version 19)
Purpose of data	Calculation of project emissions
Comments	

Data / Parameter	$\eta_{\text{COD,BL},i}$
Data unit	Percentage
Description	COD removal efficiency of the baseline system i
Source of data	Based on historical data of at least one year prior to the project implementation shall be used.
Value applied	97%
Justification of choice of data or description of measurement methods and procedures applied	
Purpose of data	For calculation of baseline emission
Comments	-

Data / Parameter	$EF_{\text{grid,CM},y}$
Data unit	tCO ₂ e/MWh
Description	Combined margin CO ₂ emission factor for grid connected power generation in year y
Source of data	Calculated as per tool 07 Tool to calculate the emission factor for an electricity system.
Value applied	0.3
Justification of choice of data or description of measurement methods and procedures applied	
Purpose of data	For calculation of baseline emission
Comments	-

Data / Parameter	$EF_{\text{CO}_2,\text{I},y}$
Data unit	tCO ₂ e/TJ
Description	Emission factor of fossil fuel replaced
Source of data	Calculated as per tool 07 Tool to calculate the emission factor for an electricity system.

Value applied	
Justification of choice of data or description of measurement methods and procedures applied	
Purpose of data	For calculation of baseline emission
Comments	-

5.2 Data and Parameters Monitored

Data / Parameter	$Q_{ww,l,y}$
Data unit	m ³
Description	Volume of wastewater treated in the project wastewater system
Source of data	On-site measurement
Description of measurement methods and procedures to be applied	Measured continuously using flow meter installed at entry of the system
Frequency of monitoring/recording	Monitored continuously (at least hourly measurements are undertaken, if less, confidence/ precision level of 90/10 shall be attained)
Value applied	used for ex ante estimations
Monitoring equipment	Flow meter
QA/QC procedures to be applied	Flow meters will undergo maintenance/calibration subject to appropriate industry standards. If local/national standards or the manufacturer's specifications are not available, international standards may be used.
Purpose of data	Calculation of baseline and project emissions
Calculation method	N/A
Comments	-

Data / Parameter	COD _{ww,untreated,y}
Data unit	t COD/m ³
Description	Chemical oxygen demand of the wastewater entering the biogas treatment system in year y
Source of data	On-site measurement
Description of measurement methods and procedures to be applied	Measured the COD according to national or international standards. COD is measured through sampling.
Frequency of monitoring/recording	Periodically, at least monthly
Value applied	used for ex ante estimations
Monitoring equipment	Laboratory analysis
QA/QC procedures to be applied	Sampling and analysis will be carried out adhering to internationally recognized procedures. Samples and measurements shall ensure a 90/10 confidence level
Purpose of data	Calculation of baseline and project emissions
Calculation method	N/A.
Comments	-

Data / Parameter	COD _{ww,treated,y}
Data unit	t COD/m ³
Description	Chemical oxygen demand of the wastewater after the treatment system affected by the project activity
Source of data	On-site measurement
Description of measurement methods and procedures to be applied	Measured the COD according to national or international standards. COD is measured through sampling
Frequency of monitoring/recording	Periodically, at least monthly

Value applied	used for ex ante estimations
Monitoring equipment	Laboratory analysis
QA/QC procedures to be applied	Sampling and analysis will be carried out adhering to internationally recognized procedures. Samples and measurements shall ensure a 90/10 confidence level
Purpose of data	Calculation of baseline and project emissions
Calculation method	N/A
Comments	-

Data / Parameter	$S_{final,PJ,y}$
Data unit	tonne
Description	Method of disposal of the sludge
Source of data	Record of site sludge disposal
Description of measurement methods and procedures to be applied	Use of sludge for land disposal is monitored.
Frequency of monitoring/recording	At each incidence of sludge disposal
Value applied	N/A
Monitoring equipment	Record
QA/QC procedures to be applied	
Purpose of data	Monitoring of sludge disposal for compliance with methodology
Calculation method	-
Comments	-

Data / Parameter	$FV_{RG,h}$
Data unit	m ³ / hour
Description	Volumetric flow of biogas recovered and flared at normal conditions in the hour h
Source of data	Measured continuously
Description of measurement methods and procedures to be applied	The volume of biogas will be measured with a normalized flow meter or calculated from the volumetric flow, pressure, temperature
Frequency of monitoring/recording	Monitored continuously (at least hourly measurements are undertaken, if less, confidence/ precision level of 90/10 shall be attained)
Value applied	m ³ / hour used for ex ante estimations
Monitoring equipment	Flow meter(s)
QA/QC procedures to be applied	-
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	$fV_{CH_4,h}$
Data unit	%
Description	Methane content in biogas in year y
Source of data	Continuous measurement
Description of measurement methods and procedures to be applied	Using calibrated gas analyzer either continuous or periodical measurement at confidence/precision level of 90/10.
Frequency of monitoring/recording	Monitored continuously (at least hourly measurements are undertaken, if less, confidence/ precision level of 90/10 shall be attained)

Value applied	used for ex ante estimations
Monitoring equipment	Gas analyzer
QA/QC procedures to be applied	-
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	T _{biogas}
Data unit	°C
Description	Temperature of biogas
Source of data	On-line monitoring
Description of measurement methods and procedures to be applied	Pressure and temperature are required to determine the density of the methane. If the biogas flow meter employed measures flow, pressure and temperature and displays/outputs normalized flow of biogas, there is no need to separate monitoring of pressure and temperature of the biogas.
Frequency of monitoring/recording	Monitored continuously (at least hourly measurements are undertaken, if less, confidence/precision level of 90/10 shall be attained)
Value applied	used for ex ante estimations
Monitoring equipment	Temperature probe
QA/QC procedures to be applied	
Purpose of data	Calculation of project emissions
Calculation method	
Comments	Pressure and temperature are required to determine the density of the methane. If the biogas flow meter employed measures flow, pressure and temperature and displays/outputs normalized flow

	of biogas, there is no need to separate monitoring of pressure and temperature of the biogas.
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Data / Parameter	P
Data unit	Pa
Description	Pressure of biogas
Source of data	On-line measurement
Description of measurement methods and procedures to be applied	Pressure measurement device.
Frequency of monitoring/recording	Monitored continuously (at least hourly measurements are undertaken, if less, confidence/precision level of 90/10 shall be attained)
Value applied	used for ex ante estimations
Monitoring equipment	Pressure measurement device.
QA/QC procedures to be applied	-
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	Flame detector
Data unit	On/off
Description	Flame detection
Source of data	On-line
Description of measurement methods and procedures to be applied	Detection of flame

Frequency of monitoring/recording	Hourly
Value applied	used for ex ante estimations
Monitoring equipment	A flame detector (thermocouple or equivalent) will be used.
QA/QC procedures to be applied	-
Purpose of data	Calculation of project emissions
Calculation method	-
Comments	For determination of flare efficiency ($\eta_{\text{flare,h}}$)

Data / Parameter	$FC_{\text{diesel,y}}$
Data unit	liter
Description	Diesel combusted in project activity in year, y
Source of data	Onsite measurement
Description of measurement methods and procedures to be applied	Use either mass or volume meters.
Frequency of monitoring/recording	Continuous monitoring and monthly compilation
Value applied	used for ex ante estimations
Monitoring equipment	Metering
QA/QC procedures to be applied	The consistency of metered fuel consumption quantities should be cross-checked by an annual energy balance that is based on purchased quantities and stock changes. Where the purchased fuel invoices can be identified specifically, the metered fuel consumption quantities should also be cross-checked with available purchase invoices from the financial records

Purpose of data	Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	WC,i,y						
Data unit	tC/mass unit of the fuel						
Description	Weighted average mass fraction of carbon in fuel type i in year y						
Source of data	The following data sources may be used if the relevant conditions apply: <table border="1"> <thead> <tr> <th>Data Source</th><th>Conditions for using the data source</th></tr> </thead> <tbody> <tr> <td>(a) Values provided by the fuel supplier in invoices</td><td>This is the preferred source</td></tr> <tr> <td>(b) Measurements by the project participants</td><td>If (a) is not available</td></tr> </tbody> </table>	Data Source	Conditions for using the data source	(a) Values provided by the fuel supplier in invoices	This is the preferred source	(b) Measurements by the project participants	If (a) is not available
Data Source	Conditions for using the data source						
(a) Values provided by the fuel supplier in invoices	This is the preferred source						
(b) Measurements by the project participants	If (a) is not available						
Description of measurement methods and procedures to be applied	Measurements should be undertaken in line with national or international fuel standards.						
Frequency of monitoring/recording	The mass fraction of carbon should be obtained for each fuel delivery, from which weighted average annual values should be calculated.						
Value applied	used for ex ante estimations						
Monitoring equipment							
QA/QC procedures to be applied	Verify if the values under (a) and (b) are within the uncertainty range of the product of the IPCC default values as provided in Table 1.2 and Table 1.3, Vol. 2 of the 2006 IPCC Guidelines. If the values fall below this range collect additional information from the testing laboratory to justify the outcome or conduct additional measurements. The laboratories in (b) should have ISO17025						

	accreditation or justify that they can comply with similar quality standards.
Calculation method	-
Comments	

Data / Parameter	$\rho_{i,y}$								
Data unit	Mass unit/volume unit								
Description	Weighted average density of fuel type i in year y								
Source of data	The following data sources may be used if the relevant conditions apply: <table border="1"> <thead> <tr> <th>Data Source</th><th>Conditions for using the data source</th></tr> </thead> <tbody> <tr> <td>(a) Values provided by the fuel supplier in invoices</td><td>This is the preferred source</td></tr> <tr> <td>(b) Measurements by the project participants</td><td>If (a) is not available</td></tr> <tr> <td>(c) Regional or national default values</td><td>If (a) is not available. These sources can only be used for liquid fuels and should be based on well-documented, reliable sources (such as national energy balances)</td></tr> </tbody> </table>	Data Source	Conditions for using the data source	(a) Values provided by the fuel supplier in invoices	This is the preferred source	(b) Measurements by the project participants	If (a) is not available	(c) Regional or national default values	If (a) is not available. These sources can only be used for liquid fuels and should be based on well-documented, reliable sources (such as national energy balances)
Data Source	Conditions for using the data source								
(a) Values provided by the fuel supplier in invoices	This is the preferred source								
(b) Measurements by the project participants	If (a) is not available								
(c) Regional or national default values	If (a) is not available. These sources can only be used for liquid fuels and should be based on well-documented, reliable sources (such as national energy balances)								
Description of measurement methods and procedures to be applied	Measurements should be undertaken in line with national or international fuel standards								
Frequency of monitoring/recording	The density of the fuel should be obtained for each fuel delivery, from which weighted average annual values should be calculated								
Value applied	used for ex ante estimations								
Monitoring equipment									
QA/QC procedures to be applied	-								
Calculation method	-								

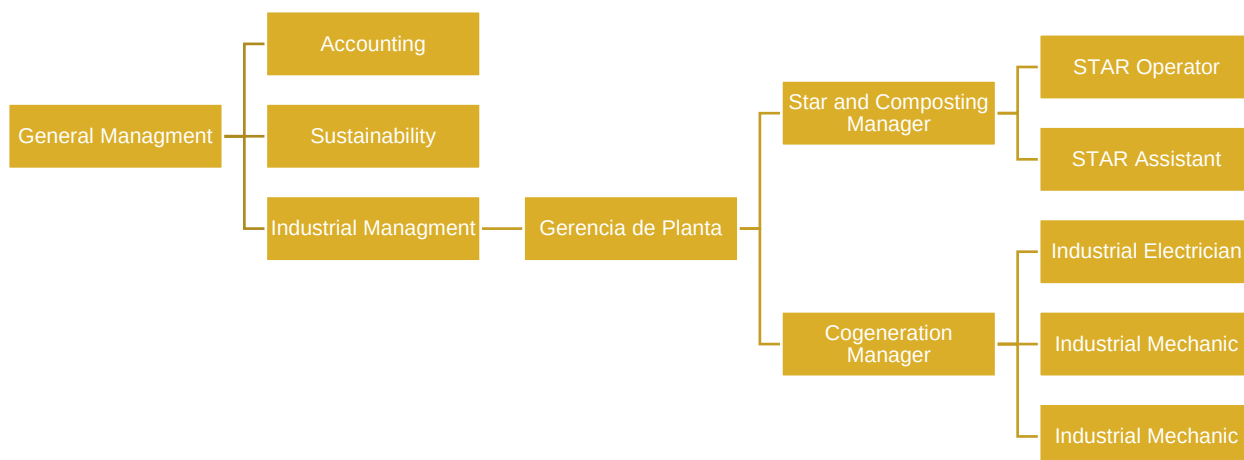
Comments

Applicable where Option A is used and where FCI.j.y is measured in a volume unit. Preferably the same data source should be used for wC,i,y and pi,y

5.3 Monitoring Plan

The project participants will monitor emission reductions by the methods, indicators, and frequency required by the VCS Standard V4.7 and methodology AMS-III.H. (Version 19.0) and AMS-I.C (Version 22.0). To ensure project ERs are measurable and real the monitoring methodology is based on direct measurement of the amount of biogas captured and destroyed by the project.

Organizational Structure Project



Responsibilities

Entity	Responsibility
Agroindustria Palmera San Roman S.A.	Operation and maintenance of monitoring equipment and infrastructure.
NEOSA	Oversight of data collection, QA/QC procedures, and preparation of monitoring reports.
External Labs / Calibration Providers	Calibration of instruments such as gas analyzers, flow meters, and pressure/temperature sensors.

QA/QC Procedures

- Calibration: All meters and sensors will be calibrated in accordance with national guidelines, annually, following manufacturer specifications or ISO guidelines. Cross-checking of internal meter readings and external sources (e.g. electricity invoices) will be completed to avoid discrepancies.
- Cross-checks: Gas production values will be cross-verified with COD balance estimates.
- Data Integrity: Redundant logging (digital + hardcopy). Automatic backups daily; reviewed monthly.
- Confidence Level: COD samples analyzed with 90/10 precision.

Data Storage and Retention

- All raw and processed data will be retained for the complete crediting period.
- In the event of a loss of electronic data, manual hard copy records will be used for estimates.
- Digital records will be stored in secured cloud servers; hard copies in designated areas.
- Data access will be granted to validation/verification bodies upon request.
- If data cannot be retrieved and manual measurements are not available, methodological alternatives to account for the gap in monitoring will be followed.

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

APPENDIX X: <TITLE OF APPENDIX>

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